

## Metaphors in AI:

Examining the language used by mathematics professors

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In October 2025, Epoch AI<sup>1</sup> conducted individual interviews<sup>2</sup> with 4 distinguished professors of mathematics from 4 different universities in California, asking them to provide their assessment and insight on the current capabilities of AI reasoning models for high-level math and scientific research. The professors, in order by date interviewed, were Greta Panova (University of Southern California), Jesus De Loera (University of California at Davis), Ravi Vakil (Stanford University), and Svetlana Jitomirskaya (University of California at Berkeley). The interviews' durations are 14, 17, 17, and 36 minutes and provide rich, meaningful, and spontaneously articulated context to analyze their viewpoints about AI, especially in cases where they use metaphors and personification to describe it.

The questions posed by Epoch AI are pertinent because in 2024, the *AlphaProof* and *AlphaGeometry 2* AI models, two new reinforcement-learning based systems for formal math reasoning and geometry built by Google's DeepMind research lab, were able to achieve the equivalent of a Silver medal at the International Mathematical Olympiad (IMO)<sup>3</sup>. This accomplishment ranks them on par with the most intellectually gifted students in the world at pre-college level.

It also serves as a pivotal checkpoint on what the current state of those models' capabilities are and what domains in reasoning they might struggle with, which was revealed to be combinatorics (as *AlphaProof* was not able to earn any points in the combinatorial mathematics problems, which comprised 2 of the 6 problems on the exam paper). This weakness has been observed in other large reasoning models (LRMs), whose pattern-recognition algorithms do not lend themselves easily to flexible (combinatorial) problem-solving, which is a weakness formally documented by Apple researchers by substituting math problems with puzzles to better isolate the AI's chain-of-thought<sup>4</sup>. Thus, the next challenge for builders of AI reasoning models would be to establish deeper capabilities for combinatorial reasoning and win the trust of the scientific research community, which relies on the very skillset to weave complex ideas together and produce novel knowledge.

While LRMs are aimed at high-level research and reasoning, i.e. AI models that mathematics professors would normally be prompting, the LLMs are the more widely-spread AI products among learners of mathematics, including college students. As such, LLM models have also been the subject of evaluation for mathematical intelligence, specifically on the *MathArena Apex*<sup>5</sup>, but, disappointingly, they had performed very poorly – the most capable LLM (Grok 4.1) had only scored 5.2%. This, however, changed with the introduction of Gemini 3 Pro by Google on November 18, 2025, because the same score for the LLM leapt from a dismal 0.5% for Gemini 2.5 Pro to 23.5% for Gemini Pro 3<sup>6</sup>. Thus, we

see a big improvement by Google's DeepMind Research Lab in making its LLM able to solve more problems rooted in mathematical reasoning.

When it comes to the capabilities of LRMs and LLMs, the same team of machine learning researchers at Apple (referenced above) conducted a side-by-side comparison and concluded that “for simpler, low-compositional problems, standard LLMs demonstrate greater efficiency and accuracy. As problem complexity moderately increases, thinking models gain an advantage. However, when problems reach high complexity with longer compositional depth, both model types experience complete performance collapse”.

Thus, it is not surprising to hear from the interviews that the overarching belief of the 4 mathematics professors is that the fledgling stage of AI's usability for higher mathematics will take time to develop with the most common answer being a timeline of at least 10 years – the complexity, esoteric obscurity of methods, and combinatorics is that high! They maintain a predominantly lukewarm sentiment towards the current abilities of LRMs (as it relates to doing advanced math and scientific research) and highlight the risk of overreliance on the more accessible GenAI forms such as ChatGPT by their student. They point that ChatGPT still makes a lot of mistakes and hallucinates on key elements in mathematical proofs and concepts. At least 2 of the professors are trying to guide their students how to properly use GenAI and avoid the trap of holding it as absolute truth, a pitfall we have seen described as the hype effect from overinflating the belief in AI's capabilities<sup>7</sup>.

It is very interesting to observe that even though the professors are shrewd and probably have the best understanding of all humans how the AI models work on technical level, all 4 make metaphorical statements about AI that anthropomorphizes those systems and brings human behavioral dimensions to the otherwise dry computational procedures behind the scenes. Anthropomorphizing AI is a trend that makes it easier to relate to the technology, but it also presents a slippery road for establishing beliefs that simply aren't there<sup>7</sup>. UNESCO has also highlighted as early as 2021 the need to exercise caution on this topic in its pioneer “Recommendation for the Ethics of Artificial Intelligence”<sup>8</sup>.

Listening to the interviews, Professor Jitomirskaya states that “... in order to solve my problem, **a machine would really need to argue** in a way that is not even written in any paper, but may be understood by some people in the field.” She says that the best approach for now is to “make it **your friend and helper**, but not an enemy of your development”. Similar to Professor Jitomirskaya, Professor De Loera upholds that “if you use it as some kind of **conversational partner**, then it's okay.” The best advice he has for his students is that “if you get stuck and you cannot do the answer, you can ask GPT. Just

check, go over the logic and try another problem. And so it can actually help you learn. It's like having **a tutor**. But if you trust it blindly, that's where it goes wrong."

In the comments made by Professor Jitomirskaya and Professor De Loera, we find a theme of camaraderie and mentorship about AI as they advise students to use GenAI tools for more common and down-to-earth activities. They show an attitude of encouragement towards the student-AI relationship but under some stern words of caution that students should remain vigilant about the outputs. These metaphors frame AI as approachable, amiable, and practical - they disarm the notion that AI will lead to a takeover in science in which independent mathematical thinking will become obsolete but suggest that GenAI can be a toolbox that can accompany the learning journey.

This placating view stands somewhat in contrast to the more frequently encountered, risk-oriented metaphors applied to AI in public policy literature<sup>9</sup> that tend to instill fear, dystopian feelings, and highly uncertain consequences for humans (metaphors such as "misinformation amplifier", "malicious actor tool", or "judge", to cite 3 examples from the author's "Function" category). Out of 55 metaphors gathered by Maas, only 2 ("companion", "advisor") align with the professors' view suggesting that perception about AI differs dramatically between the context of the classroom and the public office space.

The professors' view is in fact closer to the aggregated themes on publicly available blog posts that portray the human interaction with AI as similar to "being assisted by a guide dog" or "being assisted by a copilot while flying"<sup>10</sup>. The latter metaphor was also chosen by Microsoft when it unveiled its AI Assistant under the name of "Copilot" in March 2023 and noted that it is "the next major step in the evolution of how we interact with computing,"<sup>11</sup>. In that way, the professors' view follows the segment of society that believes in the healthy, collaborative relationship with AI because, as Ravi points out, "in the long run, we are on the same team".

By using these metaphors, the 2 professors recognize the beneficial relationship that GenAI can exert on the students who can bounce off ideas, search for examples, ask for clarifications, but they also draw a clear line that AI still cannot argue its way through an advanced-level proof and the presence of AI should not lead to the belief that studying mathematics is futile for humans – in fact all professors share the belief that mathematicians will hardly go extinct (Table 1) but with the help of AI can become more complete and innovative in their intellectual pursuits.

Table 1: Question asked to all 4 professors

| How much will AIs change math research on a scale from 0 to 10, where zero is the level of a pocket calculator and ten is human mathematicians are no longer needed? |                    |                                                                      |                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------|------------------------|
| Professor Panova                                                                                                                                                     | Professor De Lorea | Professor Vakil                                                      | Professor Jitomirskaya |
| 8                                                                                                                                                                    | 5                  | “8+3i”<br>(0-10 scale too simplistic to capture the possible effect) | 5                      |

The biggest personification of AI in the interviews comes from Professor Vakil, who is also the President of the American Mathematical Society. He says that “... **AI roughly right now is at the stage of a slightly enthusiastic but somewhat confused undergraduate.**” He also humanizes AI by linking it directly to the concept of acquiring language skills, a fundamentally innate process for development in the human brain: “I’m going to do it [use AI] like speaking a second language, whereas I expect my students to **use AI like speaking a first language...** The people born into those worlds speak them natively and then do much more powerful things with them.”

Professor Panova also likens AI to her current undergraduate students: “... you ask a question and it gives you something completely wrong. It’s like **one really unprepared but ambitious student** comes and says, I want an A. Look here, I did this. It’s not quite right, but can I have some more points? I wish it wasn’t doing this. Maybe it shouldn’t be **acting like a human** that much.” She also sees a link between AI and human-form cognition when asked to share her view about its future: “I can’t give predictions, but **it’s developing like humans** in some sense. I do expect it to be **able to do what humans can** with more computing power than we have”.

These personifications go even deeper than the previous metaphors and create an image of AI that carries human biological and psychological dimensions. The two professors imply that GenAI can be thought of as an undergraduate student, exhibiting passion, throwing tantrums, and having endless energy. This view of GenAI is again largely positive (even if Professor Panova implies that GenAI could also be annoying) because the metaphors associate GenAI with the sanguine virtues of youth, perhaps also driven by the sprightly and chipper tone that GenAI models use by default<sup>12</sup>.

We can deduce that the two professors do not recognize AI as an existential threat to human development, but rather as a high-potential student, who is developing much the same way as a college junior or senior would. However, all 4 professors point that if AI is compared to graduate students, the latter group is believed to be much more capable and

the expectations are very definitive: it may take them some time and a few hints, but graduate students would be able to arrive at complex mathematical solutions on their own with their work carrying veracity and proven techniques.

This level of capability is lacking in the performance of current AI models and all professors opine that it may be a whole decade until machines come closer to such mathematical intelligence. It is noteworthy to point that Professor Vakil makes a distinction between human intelligence at large (often referred to as AGI) and mathematical intelligence. He does not focus on the former but only tracks AI for signs that it can match the latter. His verdict so far is that “the ability to mathematically reason well is a difficult one that AIs are notoriously bad at”.

Despite this (current) shortcoming of AI in mathematical intelligence, all 4 professors believe that when applied to society at large, AI will have a tremendous effect and would accelerate its progress (Table 2).

*Table 2: Common questions to the professors*

|                                                                                                                                       | Professor Panova | Professor De Lorea | Professor Vakil | Professor Jitomirskaya |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------|-----------------|------------------------|
| How much will AI reshape the world where zero is again, a pocket calculator and ten is at least as much as the Industrial Revolution? | 9                | 7 or 8             | Not asked       | 10                     |
| If you could choose whether to stop AI progress—so that would be a zero—or accelerate as fast as possible, where would you land?      | 4 or 5           | 6 or 7             | 9               | 8                      |

In summary, throughout these 4 interviews, we hear metaphors about two sets of relations: (1) the student-to-AI relation in which students are encouraged to use GenAI as a *tutor, helper, or conversational partner* and draw a line at that and (2) the professor-to-AI relation in which professors view GenAI as resembling in ability an undergraduate student, not yet mature enough for high-level mathematical work at the frontiers of the field. Both sets of metaphors reveal a hospitable and positive attitude towards AI but with an asterisk that when it comes to mathematics, AI hasn’t mastered “the ways of thinking” yet.

Thus, it would be highly interesting to re-do these interviews after the anticipated window of 10 years when AI may have evolved and developed deeper capabilities for mathematical intelligence. At that point, the professors would also likely apply different metaphors to describe AI, some of which could involve phrases such as “peer”,

“collaborator”, “right hand”, or “trustworthy colleague”. In this future scenario (“future imaginary”), AI would then have made the leap from a Silver medal at the IMO to a fully defended Ph.D. Since this leap normally takes 9-12 years for humans to do (possible path of 4 years of undergraduate study, 2 years in a Master’s (optional) and 4-6 years in a Ph.D. program), why wouldn’t AI be able to close this gap in 10 years as well, especially since it is such an “enthusiastic student”?

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<sup>1</sup>Epoch AI is a multidisciplinary non-profit research institute investigating the future of artificial intelligence (<https://epoch.ai/>)

<sup>2</sup>Source interviews:

- October 9, 2025: Interview 1 with Greta Panova, professor of mathematics at the University of Southern California ([Why frontier AI can't solve this professor's math problem - Greta Panova - YouTube](#))
- October 13, 2025: Interview 2 with Jesus De Loera, professor of mathematics at the University of California at Davis ([Democratic proofs or fake proofs? - Jesús De Loera](#))
- October 23, 2025: Interview 3 with Ravi Vakil, professor of mathematics at Stanford University and President of the American Mathematical Society ([Waiting for AI's phase change in mathematics - Ravi Vakil - YouTube](#))
- October 27, 2025: Interview 4 with Svetlana Jitomirskaya, professor of mathematics at the University of California at Berkeley ([AI can learn logic. But can it learn folklore knowledge? - Svetlana Jitomirskaya](#))

<sup>3</sup>AlphaProof and AlphaGeometry teams, “[AI achieves silver-medal standard solving International Mathematical Olympiad problems](#)”, *Google DeepMind blog*, July 25, 2024.

<sup>4</sup>Shojaee, Parshin; Mirzadeh Iman; Alizadeh Keivan; Horton Maxwell; Bengio Samy; Farajtabar, Mehrdad, “[The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity](#)”, *Apple Machine Learning Research*, June 2025.

<sup>5</sup>[MathArena Apex](#), November 2025.

<sup>6</sup>Hassabis, Demis; Kavukcuoglu, Koray; Pichai, Sundar, “[A new era of intelligence with Gemini 3](#)”, *Google Blog*, November 18, 2025.

<sup>7</sup> Placani, Adriana, “[Anthropomorphism in AI: hype and fallacy](#)”, *AI and Ethics*, vol. 4, pp. 691-698, February 5, 2024.

<sup>8</sup> UNESCO, “[Recommendation on the Ethics of Artificial Intelligence](#)”, article #128 “Member States should implement policies to raise awareness about the anthropomorphization of AI technologies and technologies that recognize and mimic human emotions, including in the language used to mention them, and assess the manifestations, ethical implications and possible limitations of such anthropomorphization.”, November 2021.

<sup>9</sup> Maas, Matthijs, “[AI is like ... A literature review of AI metaphors and why they matter for policy](#)” [AI Foundations report 2], *Institute for Law and AI*, October 2023.

<sup>10</sup> Esbrí-Blasco, Montserrat, “[Metaphorical conceptualization of AI in digital discourse](#)”, *Methaodos: Social Science Journal*, vol 12, issue 2, 2024.

<sup>11</sup> Spataro, Jared, “[Introducing Microsoft 365 Copilot – your copilot for work](#)”, The Official Microsoft Blog, March 16, 2023.

<sup>12</sup> Mitchell, Heidi, “[That Chatbot May Just Be Telling You What You Want to Hear](#)”, *The Wall Street Journal*, June 25, 2025.